

Algorithm Design - Exam

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Instructions:

- You are given the question sheet and 4 sheets for your answers. The problems are 4.
- Before you start, write on the top of each sheet (1) your full name, (2) your student ID (*matricola*) and (3) the sheet number (1, 2, 3, and 4).
- If you use additional sheets as rough draft (*brutta*) for calculations, etc., you must hand it, along with the rest of the exam. You can keep the question sheet.
- When your pages are full, you can ask for extra pages; make sure that you also write there your name, student id, and the sheet number (5, 6, etc.). You must hand all the sheets that you receive.
- The rooms are small so it may be tempting to copy. Note that if we catch you copying in any way (copying from another student, using your phone, etc.) you will be automatically disqualified.

Problem 1

You have decided to drive from Rome to Paris for winter break and you want to figure out which hotels you should stay at along the way. Assume that there is only one highway from Rome to Paris, that it is exactly n miles long and that there is exactly one hotel at each mile along this highway. Each day, you can drive at most k miles and then, if you are not yet at Paris, you must stop at a hotel for the night. You have as **input** an array A such that for each $i < n$, $A[i]$ contains the cost to stay for one night at the hotel at mile i along the highway. The goal is to find the minimum possible total **amount of money** c that you can spend on hotels during your trip.

Example. Suppose $n = 10$, $k = 3$ and the costs of the hotels, in order, are 3, 1, 4, 5, 2, 3, 7, 8, 1 (note that there are only 9 hotels because after you have driven 10 miles, you have reached Paris and do not need a hotel). Then the minimum total cost is 7, which can be obtained by staying at the hotels at miles 2, 5, 6 and 9.

1. Describe a dynamic programming solution with subproblems $C(i)$ defined as the minimum cost to travel the first i miles.
2. Provide the pseudocode of the algorithm that finds the value of the optimal solution c .
3. Analyze the running time of the algorithm at point 2.

Problem 2

Let $G = (V, E)$ be a directed graph and for every edge $e \in E$ let $c(e) \in \mathbb{N}$ be its *integer* non-negative capacity. Let $s, t \in V$ be a source and a sink node let $f : E \rightarrow \mathbb{N}$ be a *max flow* from s to t and assume f is *acyclic*, i.e., there is no cycle in G in which all edges have positive flow.

Suppose the capacity of one specific edge $\hat{e} \in E$ decreases of one unit, i.e., consider the new capacities

$$\hat{c}(e) = \begin{cases} c(e) - 1 & \text{for } e = \hat{e} \\ c(e) & \text{for } e \neq \hat{e} \end{cases}$$

Design an algorithm that, starting from f , finds a max flow from s to t in the graph G with the new capacities $\hat{c}$ in time $\mathcal{O}(|V| + |E|)$.

Hint: formulate the problem in the residual capacity network.

Problem 3

Consider the following problems:

- **k-coloring:** given a graph $G(V, E)$ and an integer $k < |V|$, can you color the vertices V with k colors such that no two adjacent vertices have the same color?
- **Class scheduling:** Given a set n of classes, each with a time slot S , and an integer $k < n$, can you assign the classes to k rooms such that no class has an overlapping time slot?

Show that Class scheduling is NP-complete using a reduction from k -coloring.

Problem 4

1. Define the k -center problem in a metric space.
2. Show that the k -center problem is NP-hard.
3. Give a simple 2-approximation algorithm for k -center and give the proof for the approximation result.